

Call Center Waiting Line System

Experimental Report: Priority Queue Design and Instant-Pickup Verification

Group 7 — CSD201

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1 Introduction

This report presents empirical results from the *Call Center Waiting Line System* simulation, a multi-threaded Java program modeling a call center with a Dispatcher thread, multiple Agent threads, and an Aging thread. Two experimental questions are addressed:

1. **Experiment 1:** Does a Dual Queue design (separate VIP / Regular queues) or a Single Queue with an Aging mechanism produce a lower Average Wait Time (AWT) for Regular (non-VIP) customers?
2. **Experiment 1B:** Verification of the instructor’s hypothesis — when all 10 agents are busy and a new call arrives, is it queued rather than answered instantly? Conversely, are calls answered immediately only when at least one agent is free at arrival time?

Each experiment was executed twice (independent runs) to check the stability and reproducibility of the results.

2 Experimental Setup

Parameter	Value
Number of calls (Experiment 1)	500 (fixed, shared by both scenarios)
Number of calls (Experiment 1B)	250 (fixed, separate dataset)
VIP ratio	20%
Number of Agents	10
Handling time per call	30–180 s (real-world seconds)
Processing speedup factor	100× real time
Instant-pickup threshold	pickup delay < 0.05 s

Table 1: Simulation configuration shared across both experimental runs.

3 Experiment 1: Dual Queue vs. Single Queue with Aging

3.1 Results — Run 1

Run 1 improvement (Scenario B relative to A, positive = B better for Regular customers): AWT reduced by -0.44% (i.e. slightly worse), Max WT reduced by 0.91% .

Metric	Scenario A (Dual Queue)	Scenario B (Aging Queue)
Regular AWT	18 min 37.7 s (1117.7 s)	18 min 42.7 s (1122.7 s)
VIP AWT	12.4 s	11.0 s
Overall AWT	15 min 27.6 s (927.6 s)	15 min 31.5 s (931.5 s)
Regular Max WT	30 min 04.1 s (1804.1 s)	29 min 47.6 s (1787.6 s)
VIP Max WT	30.8 s	47.1 s

Table 2: Experiment 1 – Run 1 results (500 calls, 20% VIP, 10 agents).

Metric	Scenario A (Dual Queue)	Scenario B (Aging Queue)
Regular AWT	14 min 35.1 s (875.1 s)	14 min 35.4 s (875.4 s)
VIP AWT	10.0 s	8.6 s
Overall AWT	11 min 49.0 s (709.0 s)	11 min 48.9 s (708.9 s)
Regular Max WT	24 min 31.1 s (1471.1 s)	24 min 26.9 s (1466.9 s)
VIP Max WT	1 min 06.1 s (66.1 s)	1 min 09.6 s (69.6 s)

Table 3: Experiment 1 – Run 2 results (500 calls, 20% VIP, 10 agents).

3.2 Results — Run 2

Run 2 improvement (Scenario B relative to A): AWT reduced by -0.02% , Max WT reduced by 0.29% .

3.3 Analysis

Across both independent runs, the Regular-customer Average Wait Time is virtually identical between the two scenarios, differing by less than half a second in both cases (Run 1: 1117.7 s vs. 1122.7 s; Run 2: 875.1 s vs. 875.4 s), corresponding to a relative difference of well under 1%. Scenario B (Aging) shows a marginally *lower* maximum wait time for Regular customers in both runs, but the effect is small and within normal run-to-run variability, as shown by the fact that Run 1 and Run 2 themselves differ by several minutes in absolute AWT due to random call-arrival and handling-time variation.

For VIP customers, Scenario B (Aging) consistently produces a slightly *lower* AWT than Scenario A (Dual Queue) in both runs (Run 1: 11.0 s vs. 12.4 s; Run 2: 8.6 s vs. 10.0 s), while the VIP maximum wait time is slightly *higher* under Scenario B in both runs (Run 1: 47.1 s vs. 30.8 s; Run 2: 69.6 s vs. 66.1 s). This is an expected trade-off of aging-based prioritization: VIP calls are still prioritized on average, but a VIP call can occasionally accumulate more wait before its aging-adjusted priority overtakes the queue, producing a higher tail (max) wait than a strict, always-VIP-first dual queue would.

3.4 Conclusion for Experiment 1

Under the tested configuration (500 calls, 20% VIP, 10 agents), neither queueing strategy provides a practically significant reduction in Regular customer Average Wait Time; the two scenarios are effectively equivalent for Regular customers (difference $< 0.5\%$ across both runs), while Scenario B (Single Queue with Aging) yields marginally better VIP average wait at the cost of a marginally higher VIP worst-case wait. This suggests that with only 10 agents serving a load of 500 calls at 20% VIP, the system is heavily saturated (see Experiment 1B below), and queue discipline alone cannot compensate for insufficient agent capacity – both designs converge to similar Regular-customer wait times because Regular customers are, in practice, served only once VIP and aged calls are cleared, regardless of the specific queueing structure used.

4 Experiment 1B: Instant Pickup Verification

4.1 Hypothesis

The instructor’s hypothesis states that whether an incoming call is answered immediately depends on agent availability at the moment of arrival: if at least one of the 10 agents is idle when a call arrives, that call is picked up instantly (pickup delay ≈ 0 s); if all 10 agents are busy handling other calls, the new call must wait in queue, and pickup delay will be strictly greater than zero.

4.2 Results — Run 1

Metric	Group A: Instant Pickup	Group B: Queued / Delayed
Condition	pickup delay < 0.05 s	pickup delay ≥ 0.05 s
Meaning	Agent available on arrival	All agents busy on arrival
Calls (out of 250)	12	238
Ratio	4.8%	95.2%
Average pickup delay	0.0000 s	7 min 26.8 s (446.8 s)
Minimum pickup delay	0.0000 s	1.5 s
Maximum pickup delay	0.0000 s	14 min 39.9 s (879.9 s)

Table 4: Experiment 1B – Run 1 results (250 calls, 10 agents).

4.3 Results — Run 2

Metric	Group A: Instant Pickup	Group B: Queued / Delayed
Condition	pickup delay < 0.05 s	pickup delay ≥ 0.05 s
Meaning	Agent available on arrival	All agents busy on arrival
Calls (out of 250)	10	240
Ratio	4.0%	96.0%
Average pickup delay	0.0000 s	7 min 03.4 s (423.4 s)
Minimum pickup delay	0.0000 s	50.0 s
Maximum pickup delay	0.0000 s	15 min 03.0 s (903.0 s)

Table 5: Experiment 1B – Run 2 results (250 calls, 10 agents).

4.4 Analysis

Both runs confirm the hypothesis with high consistency:

- In Run 1, only 12/250 calls (4.8%) were answered instantly (average and maximum delay measured at 0.0000 s, i.e. within the 0.05 s threshold), corresponding to the cases where at least one agent was idle at the moment of call arrival.
- In Run 2, the instant-pickup group was 10/250 calls (4.0%), with the same 0.0000 s delay characteristic.
- In both runs, the remaining ~ 95 – 96% of calls (238/250 and 240/250, respectively) experienced a strictly positive pickup delay, with minimum delays of 1.5 s (Run 1) and 50.0 s (Run 2) – confirming that once all 10 agents are occupied, any newly arriving call is forced to wait, with no exception observed.

- The average delay for the queued group is substantial in both runs (7 min 26.8 s and 7 min 03.4 s respectively), and the maximum observed delay reaches nearly 15 minutes in both runs, illustrating that under this call volume and agent count, queuing delay is not a rare edge case but the dominant behavior of the system.

The sharp binary separation between the two groups – a pickup delay of exactly 0.0000 s for the "instant" group versus a delay of at least 1.5 s (and averaging several minutes) for the "queued" group – with no calls observed in an intermediate zone, is strong evidence that the simulation correctly implements the intended behavior: an agent picks up a call immediately if and only if an agent is idle at arrival time; otherwise, the call is placed into the queue and only picked up once an agent becomes free.

4.5 Conclusion for Experiment 1B

The instructor's hypothesis is verified by both experimental runs. Only a small minority of calls (4.0–4.8%) arrive when an agent is free and are answered with effectively zero delay, while the large majority ($\geq 95\%$) arrive when all 10 agents are occupied and must wait in queue, with wait times ranging from a few seconds up to roughly 15 minutes. This confirms that, at the tested load (250 calls across 10 agents with 30–180 s handling times), the system operates in a persistently saturated state, and instant pickup is the exception rather than the rule.

5 Overall Conclusions

1. **Queueing strategy (Experiment 1):** For Regular customers, Dual Queue and Single Queue with Aging produce statistically indistinguishable Average Wait Times under the tested load (20% VIP, 10 agents, 500 calls); the Aging mechanism offers a small AWT benefit to VIP customers at the cost of a slightly higher VIP worst-case wait, without materially helping or harming Regular customers.
2. **Instant pickup hypothesis (Experiment 1B):** Confirmed – calls are answered instantly only when an agent is idle upon arrival (4.0–4.8% of calls across both runs), while all other calls (95.2–96.0%) must wait, with substantial average and maximum delays. This demonstrates that the 10-agent configuration is under-provisioned relative to the offered call load, which is also the underlying reason Experiment 1's two queueing strategies converge to similar Regular-customer outcomes: capacity, not queue discipline, is the binding constraint.